

PhysMeter: Benchmarking Physical Consistency in Video World Models

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Video world models can produce visually convincing yet physically inconsistent sequences, potentially limiting their reliability as world models for prediction and planning. Existing evaluations often rely on model-based judgments or reference videos, while direct physical tests largely focus on mechanics. We introduce **PhysMeter**, a measurement-based benchmark for evaluating physical consistency in video world models. PhysMeter comprises 40 controlled tasks spanning mechanics, optics, fluids, thermal and phase-change phenomena, electromagnetism, and surface tension. Each task pairs an initial image and a generation prompt with predefined physical criteria, enabling interpretable tests of observable relationships without requiring reference videos. Its evaluator combines task-observability screening with quantitative physical measurements. Experiments on eight video generation models across 1,280 videos reveal persistent physical inconsistencies and substantial variation across tasks, with the best model achieving an overall score of 57.90 out of 100. Evaluation on synthetic videos with known physical relationships supports the measurement module under controlled conditions. Our evaluator also achieves higher agreement with human judgments than a direct vision-language model baseline in both within-task rankings and pairwise comparisons. By grounding physical scores in measurable evidence and making measurement limitations explicit, PhysMeter provides an interpretable framework for diagnosing model weaknesses and assessing progress toward physically consistent video world models.

[**Kaisen**: Suggested opening (two backgrounds in two sentences): “Video generation models produce visually convincing sequences that violate physical constraints, limiting their reliability as world models for prediction and planning. Existing evaluations often rely on learned judgments or reference videos, while direct physical tests largely focus on mechanics.” Keep the current result values for the next data update.]

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1 Introduction

Using video generation models as world models for prediction and planning requires them to reproduce how environments evolve under physical constraints. Despite substantial advances in visual quality, temporal coherence, and scene complexity (Bar-Tal et al., 2024; Yang et al., 2025; Polyak et al., 2024; Kong et al., 2024; Wan et al., 2025; Zheng et al., 2025; Ma et al., 2026; Seedance et al., 2026; Team et al., 2025), current models can still generate physically implausible motions and interactions (Bansal et al., 2025; Meng et al., 2024; Rädtsch et al., 2026). As video models are increasingly explored as learned world simulators that predict how environments change in response to actions (Yang et al., 2023; Bruce et al., 2024; Qian et al., 2026; Zhao et al., 2026), such violations may undermine the reliability of simulated outcomes and the plans derived from them. Assessing their suitability for this role therefore requires systematic evaluation of whether generated events satisfy the relevant physical relationships. This evaluation should identify which phenomena models reproduce reliably, characterize where they fail, and determine whether new methods reduce these failures, providing a basis for measuring progress toward physically consistent video world models.

Existing evaluations of physical consistency draw on vision-language model (VLM) judgments, reference-based comparisons, or direct tests of physical relationships. VLM-based methods cover diverse phenomena, but their judgments depend on the evaluator’s own physical reasoning (Bansal et al., 2025; Meng et al., 2024; Lin et al.,

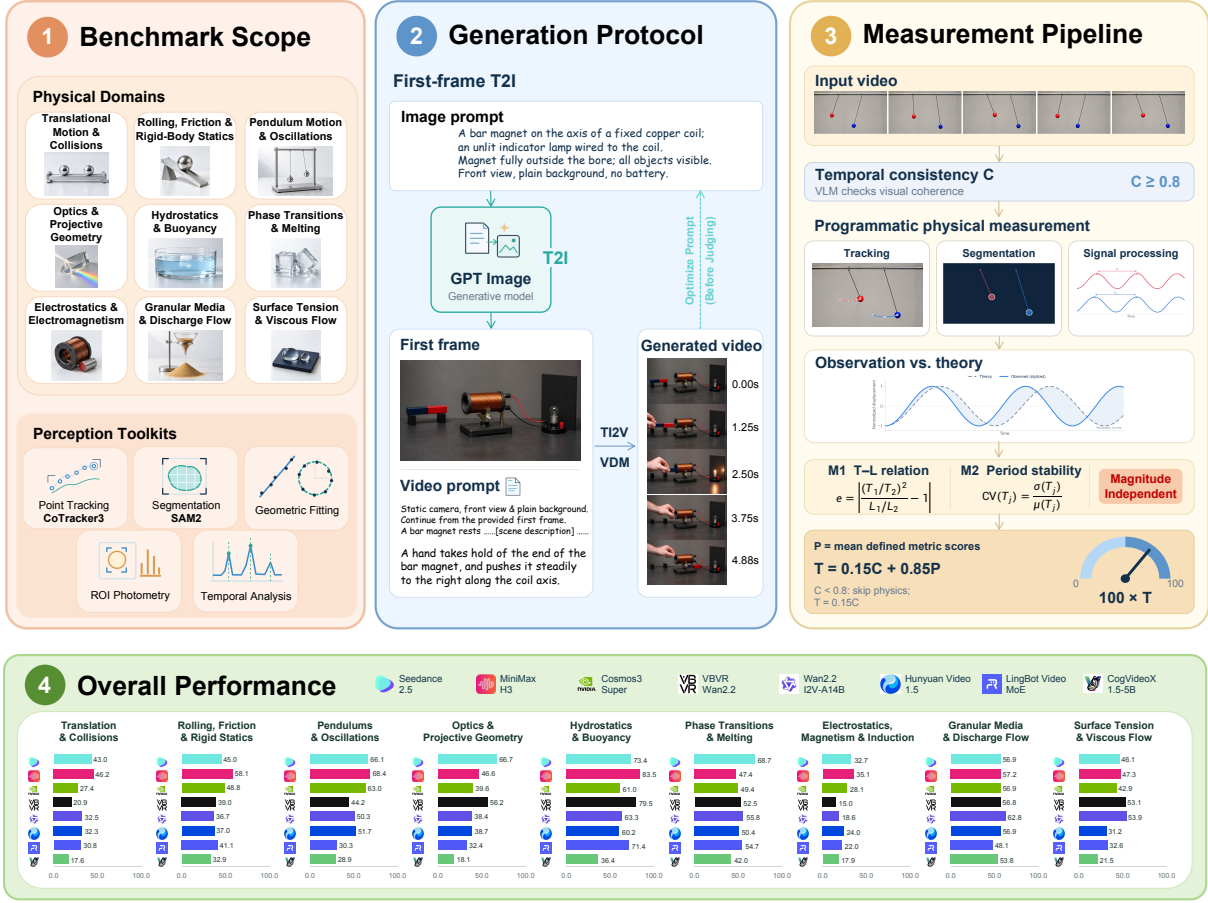


Figure 1 Overview of PhysMeter. (1) **Benchmark scope:** 40 controlled tasks span nine physical categories, supported by perception tools for extracting observable quantities. (2) **Generation protocol:** Each task pairs a generated initial frame with a task-specific video prompt, refined during benchmark construction. The coil-and-magnet scene illustrates this process. (3) **Measurement pipeline:** A separate pendulum example illustrates consistency screening, physical measurement, and comparison with predefined physical criteria. (4) **Overall performance:** Mean composite scores summarize the performance of eight video generation models across the nine categories. Normalized composite scores are multiplied by 100 for presentation, with higher values indicating better performance.

2026; Luo et al., 2026). Current VLMs still have limited capabilities in understanding physical relationships and reasoning about dynamic interactions, which can compromise the reliability of their evaluations. For example, PQSG reports that GPT-5.5 achieves 64.6% accuracy on physics questions in FinePhyEval, compared with 88.4% on object-level questions, illustrating the difficulty of judging physical behavior (Pothiraj et al., 2026). Reference-based approaches assess generated videos through comparisons with reference observations. Physics-IQ covers multiple physical domains, but evaluates similarity to reference videos rather than directly testing physical laws, making its scores sensitive to reference quality and video alignment (Rädsch et al., 2026). Direct physical tests, including Principia, instead measure quantities such as object motion and assess whether they satisfy physical laws or the conditions specified in the prompt (Thozhiyoor et al., 2026a; Le et al., 2025; Khanbayov and Kurban, 2026; Thozhiyoor et al., 2026b). These tests mainly cover mechanics, while some measurement-based approaches also require real recordings, calibrated data, or simulator ground truth (Wang et al., 2026; Jain and Wu, 2026). These gaps motivate a framework for testing observable physical relationships across domains, in which physical scores are derived from task-specific measurements without requiring reference videos or simulator ground truth, while a VLM provides preliminary consistency and observability screening.

To address this challenge, we introduce **PhysMeter**, a measurement-based benchmark comprising 40 controlled

tasks spanning mechanics, optics, fluids, thermal and phase-change phenomena, electromagnetism, and surface-tension effects. Our central idea is to pair controlled generation conditions with measurable physical criteria. As shown in Figure 1, each task defines an image-prompt input, a target physical process, and observable relationships for evaluating the generated video. PhysMeter first screens videos for temporal consistency and task observability. For videos that pass, it uses tracking, segmentation, and temporal analysis to extract task-specific quantities, such as oscillation periods, reflection angles, and liquid levels. These measurements are compared with predefined physical criteria, distinguishing measured violations from cases with insufficient evidence. The resulting composite scores combine screening and physical measurement results to summarize model performance across the benchmark’s nine task categories.

We evaluate eight video generation models on 40 tasks, collecting 1,280 generated videos. Although 98.36% of the videos pass the temporal-consistency check, their performance on physical tests remains limited. Sufficient measurement evidence is available for 716 videos, of which 159 (22.21%) pass all required physical checks. These results show that temporal coherence alone does not establish physical consistency. They also highlight the importance of reporting measurement coverage alongside physical performance, since videos with insufficient evidence cannot be reliably classified as physically correct or incorrect.

Our contributions are threefold:

- We introduce **PhysScope**, a benchmark of 40 controlled video generation tasks covering mechanics, optics, fluid behavior, thermal and phase-change phenomena, electromagnetism, and surface-tension effects.
- We develop an interpretable evaluation protocol that combines temporal-consistency screening with task-specific physical measurements and separately reports physical test outcomes and insufficient measurement evidence.
- We conduct a systematic evaluation of eight video generation models across 1,278 videos, revealing a gap between temporal coherence and performance on explicit physical tests.

2 Method

2.1 Overview and Task Formulation

PhysScope evaluates the physical accuracy of generated videos through controlled experiments and quantitative measurements. It checks whether measurable quantities in the videos, such as an object’s position and speed, follow the physical relationships expected under the specified experimental conditions.

[Kaisen: Consider defining the title’s “proxy-based” scope here: “PhysScope tests observable physical relationships under controlled experimental conditions. Positions, periods, and angles serve as measurable proxies for physical behavior.” State that these tests assess the specified relationships under task assumptions, leaving unmeasured properties unverified.]

A task specifies physical assumptions $\mathcal{A}_k$, an image-prompt instance $(I_{0,k}, p_k)$, and measurable criteria $\mathcal{R}_k$:

$$\mathcal{T}_k = (\mathcal{A}_k, I_{0,k}, p_k, \mathcal{R}_k), \quad V_{k,r} = G_\theta(I_{0,k}, p_k), \quad (1)$$

where G_θ is the evaluated video model and $V_{k,r}$ is the r -th video generated for task k . The assumptions describe materials, initial conditions, and the observation geometry; the criteria identify the physical relationships or events to evaluate. For example, a free-fall task assumes that an object is released from rest with negligible air resistance, then checks whether the measured falling distance is proportional to the square of elapsed time.

As shown in Figure 1, PhysScope evaluates video models across nine physical domains through a standardized generation and measurement workflow. For each task, an initial frame is generated using GPT-Image-2.5 () and paired with a video prompt specifying how the experiment should proceed. The video model uses these inputs to generate the experimental process. Each generated video is first checked for temporal consistency and semantic coherence; videos that meet the consistency threshold proceed to programmatic physical measurement using tools including CoTracker3 ? for point tracking, SAM 2 ? for segmentation, and task-specific routines for geometric fitting, region-of-interest photometry, and temporal analysis. Measurements extracted from the

Table 1 The nine task categories, representative phenomena, and governing physical principles.

Task category	Representative phenomena	Physical principles
Translational Motion and Collisions	Free fall, projectiles, bouncing, and collisions	Constant gravitational acceleration, ballistic motion, restitution, and momentum conservation
Rolling, Friction, and Rigid-Body Statics	Rolling descent, sliding, tipping, and hanging chains	Rolling constraints, rotational inertia, Coulomb friction, torque balance, and catenary equilibrium
Pendulum Motion and Oscillations	Period dependence on amplitude, mass, and length	Small-angle isochronism, mass independence, length scaling, and nonlinear pendulum dynamics
Optics and Projective Geometry	Reflection, refraction, shadows, and marked-rod motion	Reflection law, Snell’s law, critical-angle condition, ray concurrency, and cross-ratio invariance
Hydrostatics and Buoyancy	Liquid surfaces, communicating vessels, and floating ice	Hydrostatic pressure balance, free-surface orientation, and Archimedes’ principle
Phase Transitions and Melting	Ice melting, freezing expansion, and comparative melting	Mass conservation, buoyancy, density-dependent volume changes, and heat transfer
Electrostatics, Magnetism, and Electromagnetic Induction	Charge repulsion, compass deflection, induction, and magnetic damping	Electrostatic force balance, current-generated magnetic fields, Faraday’s law, and Lenz’s law
Granular Media and Discharge Flow	Sandpile geometry and sand–water discharge comparisons	Angle-of-repose consistency, Beverloo discharge scaling, and Torricelli’s law
Surface Tension and Viscous Flow	Capillary rise, connected bubbles, droplet merging, and viscous settling	Jurin’s law, Young–Laplace pressure, volume conservation, and Stokes drag

generated video itself are compared with the physical relationships or behaviors specified by the task. The resulting physical scores are combined with consistency scores and aggregated to compare model performance across domains.

2.2 Physics-Grounded Task Design

Task coverage and design principle. We construct 40 tasks organized into nine categories, covering a broad range of common physical phenomena encountered in everyday life. Table 1 summarizes representative phenomena and their governing physical principles. Our central design principle is to formulate physical predictions as observable relationships that can be tested without absolute spatial or temporal calibration. We organize these tests into three types: spatial relationships among geometric quantities, temporal relationships among periods, durations, or event timings, and coupled spatiotemporal relationships between motion and geometry. For each task, we identify the governing law and its assumptions, derive the relationship to be tested, and design a scene that makes the required observables measurable within a single video. Where appropriate, matched comparative setups allow shared unknown factors to cancel from the tested relationship.

Spatial relationships. Spatial tests check whether lengths, angles, or shapes agree with physical predictions. Consider two capillary tubes containing the same liquid under identical wetting conditions. A narrower tube should produce a higher liquid column: halving the tube radius should double the rise height. Let r_1, r_2 denote the tube radii and h_1, h_2 the rise heights above the reservoir surface. We compare the measured height ratio h_1/h_2 with the predicted ratio r_2/r_1 :

$$r_{\text{spatial}} = \left| \frac{h_1/h_2}{r_2/r_1} - 1 \right|. \quad (2)$$

A residual near zero indicates agreement with the prediction. All lengths can be measured in pixels under a common spatial scale, since taking ratios removes the need to convert pixels into physical distances.

[Kaisen: Add the observation condition to the scale-cancellation claim: “A common spatial scale lets us measure length ratios in pixels. This cancellation requires appropriate viewing geometry and does not remove arbitrary perspective distortion.” This would bring the caveat already in the task appendix into the main method.]

Temporal relationships. Temporal tests examine whether measured periods, durations, or event timings satisfy the expected physical relationship. For example, two small-angle pendulums with equal lengths under the same gravity should have approximately equal periods, even when their bob masses differ. We measure their periods T_1 and T_2 in frames and evaluate

$$r_{\text{temporal}} = \left| \frac{T_1}{T_2} - 1 \right|. \quad (3)$$

This test checks whether changing the bob mass preserves the oscillation period. Because both periods share the same temporal scale, the unknown conversion from frames to physical time cancels.

Coupled spatiotemporal relationships. Spatiotemporal tests examine whether motion measurements are consistent with object geometry. For example, pure rolling without slipping requires the translational speed v to equal the angular speed ω multiplied by the radius R . We measure v in pixels per frame, ω in radians per frame, and R in pixels, and evaluate

$$r_{\text{spatiotemporal}} = \left| \frac{v}{\omega R} - 1 \right|. \quad (4)$$

This test checks whether the observed translation matches the rotation of a visible surface marker. Because v and ωR share the same spatial and temporal scale factors, both unknown conversions cancel.

2.3 First-Frame Construction and Video Generation

Reliable quantitative evaluation requires clearly defined experimental conditions and observable quantities for measurement. Ambiguous initial geometry, occlusion, or an incomplete observation interval can make a physical test inconclusive and confound model errors with limitations of the input setup. We therefore construct task-specific first frames and video prompts through an iterative process, as illustrated in Figure 1, to establish the prescribed initial configuration and make the target process observable.

For each task, we translate the experimental requirements into an image prompt and use GPT Image to generate candidate first frames that establish the initial configuration and expose the quantities needed for measurement. Each first frame is paired with a video prompt specifying how the experiment begins and unfolds, which scene properties remain fixed, and the observation interval required for evaluation. For example, pendulum experiments require fixed pivots and sufficient oscillations to estimate periods, while melting experiments require visible initial and final liquid levels. We iteratively refine both prompts through pilot generation and human inspection: ambiguous configurations or poorly visible objects lead to image-prompt revisions and regenerated first frames, while unclear actions or insufficient observation intervals lead to video-prompt revisions. Revised pairs are tested again, forming a feedback loop between input construction and pilot inspection. Errors in extracting clearly visible evidence are addressed separately through evaluator review. The finalized image-prompt pairs are then submitted to eight video models through a common generation interface. The evaluation covers 40 tasks across nine categories, yielding 1,280 videos with four samples per task-model pair.

2.4 Physical Evaluation Protocol

Generated videos do not always provide the evidence required for a meaningful physical test. For example, estimating free-fall acceleration requires a consistently identifiable ball and a recoverable trajectory. If the ball disappears, changes identity, or moves discontinuously in a way that prevents reliable tracking, the required measurements become unavailable. We therefore assess measurement feasibility before testing physical

alignment. As illustrated in Figure 1, we use Qwen3.6-27B (?) to assess temporal consistency $C(V) \in [0, 100]$, including object persistence and visual continuity, as an initial screening signal. Videos with $C(V) < 80$ do not proceed to physical evaluation. This initial check assesses whether reliable measurements can be obtained; measurable behavior proceeds to the physical tests even if it is physically incorrect.

[Kaisen: Screening and recoverable measurements are different conditions. Consider replacing “This initial check assesses whether reliable measurements can be obtained” with “This screen flags visual discontinuities that can obstruct measurement; passing it does not guarantee recovery of every required quantity.” Keep physically incorrect but measurable behavior eligible for the physical tests.]

Temporal consistency alone does not establish physical correctness: a clearly visible, continuously tracked object can still follow an incorrect trajectory or violate a motion–geometry relationship. We therefore compare recovered observables with task-specific physical predictions. For each task k , an extractor E_k uses segmentation, tracking, geometric fitting, and event detection to recover quantities such as trajectories, periods, ray angles, contours, and liquid levels over the prescribed observation interval. For each indicator, we compute

$$\mathbf{z}_k = E_k(V), \quad r_{k,j} = f_{k,j}(\mathbf{z}_k; \mathcal{A}_k), \quad q_{k,j} = \phi_{k,j}(r_{k,j}) \in [0, 1], \quad (5)$$

where $\mathcal{A}_k$ denotes the experimental assumptions, $f_{k,j}$ computes a physical residual or event statistic, and $\phi_{k,j}$ maps it to an agreement score. Larger residuals receive lower agreement, while directional and event-based indicators follow their corresponding definitions. The primary indicator evaluates the target relationship, and auxiliary indicators provide complementary checks. After task-specific handling of unavailable measurements, the indicators are combined into a physics score:

$$P_k(V) = 100 F_k(\mathbf{q}_k), \quad \mathbf{q}_k = (q_{k,j})_{j \in \mathcal{J}_k}, \quad (6)$$

where $\mathcal{J}_k$ contains the task’s indicators and F_k is its aggregation rule. Intermediate measurements and visual evidence are retained to help distinguish extraction errors from physical violations.

Reporting physical agreement only for measurable videos would omit outputs that fail the initial screening and could overstate a model’s overall performance. We therefore combine temporal consistency and physical agreement into a total score that includes every available video:

$$T_k(V) = 0.15C(V) + 0.85\tilde{P}_k(V), \quad \tilde{P}_k(V) = \begin{cases} P_k(V), & C(V) \geq 80 \text{ and a physics score is recorded,} \\ 0, & \text{otherwise.} \end{cases} \quad (7)$$

Physical agreement receives the larger weight, while videos failing the screening receive only the temporal-consistency contribution. For each model, temporal consistency and total scores are averaged over all available videos. We average all recorded physics scores, including zeros, for each model and each category or difficulty level.

3 Experiments

3.1 Settings

Video Generation Models. We evaluate eight image-to-video models: CogVideoX1.5-5B, Cosmos3-Super, HunyuanVideo-1.5, LingBot-Video, MiniMax-H3, Seedance-2.5, VBVR-Wan2.2, and Wan2.2-I2V-A14B (?). For each of the 40 tasks, all models receive the same initial frame and task description. We generate four samples per model–task pair, giving a planned total of 1,280 videos. Random seeds are fixed to 42–45 where supported. Model-specific generation settings are documented for reproducibility.

Metrics. Each video is evaluated for task consistency and physical correctness. The consistency evaluator uses a dedicated prompt for each task to assess whether the required subjects, events, and comparisons are observable. Task-specific physical evaluators extract trajectories, periods, angles, and event timings using color segmentation, connected-component analysis, optical flow, and contour or line fitting, with SAM2 (?) and CoTracker (?) providing object segmentation and point tracking where needed. The extracted measurements

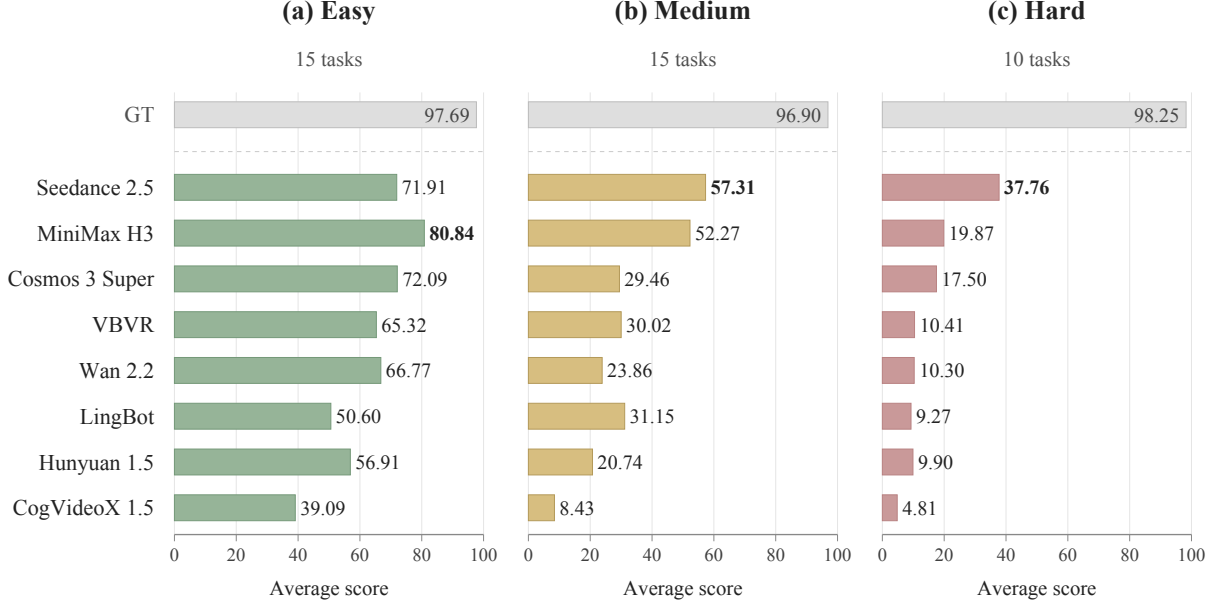


Figure 2 Performance of each video generation model across Easy, Medium, and Hard physical phenomena. Colored bars show reviewed composite scores, while light-gray bars show physical scores on synthetic GT videos with the consistency gate bypassed. All scores are scaled to 0–100. The best generation model for each difficulty level is shown in bold.

are compared with the corresponding physical constraints to compute task-specific errors and scores. Most residual-based metrics map an error e to a score through

$$s(e; a) = \frac{1}{1 + |e|/a}, \quad (8)$$

where $a > 0$ is a predefined, metric-specific error scale. For example, the pendulum-length task uses $a = 0.20$ and computes

$$e = \left| \frac{(T_1/T_2)^2}{L_1/L_2} - 1 \right|, \quad (9)$$

where T_1, T_2 and L_1, L_2 are the periods and lengths of the two pendulums, respectively. Other indicators use task-specific mappings, including event F1 and binary directional agreement. The physical score is the arithmetic mean of the indicators defined for each task. All normalization parameters remain fixed across generative models.

3.2 Main Results

Results by Task. Table 2 suggests that physical understanding varies substantially across different models and remains limited overall. Even the strongest model achieves an average score of only 57.90, indicating considerable room for improvement in consistently reproducing physical behavior. Three tasks illustrate these findings. Melting ice containing a stone (P21b) is challenging for every evaluated model, with a model-averaged score of 3.45 and a maximum of 15.00. The uniformly low scores reveal a shared weakness in this melting and buoyancy scenario, rather than a limitation confined to a particular model. Charged-sphere equilibrium (P28) reveals a substantial difference between models: Seedance and MiniMax score 98.34 and 97.24, respectively, whereas all other models score at most 15.00. This gap suggests that the ability to reproduce electrostatic equilibrium varies markedly across models, with strong performance concentrated in only two of the evaluated systems. Communicating-vessel equilibrium (P19), however, provides an example of consistently strong performance. All eight models score between 93.09 and 99.01 and pass the consistency check on every available video, demonstrating broad success in reproducing this particular equilibrium condition.

Task			Level	 Seedance	 MiniMax	 Cosmos	 VBVR	 Wan	 LingBot	 Hunyuan	 CogVideoX
				2.5	H3	3 Super	Wan2.2	2.2-A14B	30B-A3B	1.5	1.5-5B
1. Translational Motion and Collisions											
P1	Free fall	M		29.50 ✓	25.84 ✓	34.54 ✓	24.27 ✓	22.72 ✓	34.41 ✓	23.60 ✓	18.03 ✓
P2	Projectile motion	H		29.13 ✓	39.01 ✓	17.25 ✗	27.64 ✓	10.36 ✗	15.43 ✗	7.83 ✗	0.00 ✗
P3	Complementary-angle throws	M		53.96 ✓	57.66 ✓	20.97 ✓	10.88 ✗	0.00 ✗	30.87 ✗	3.38 ✗	0.00 ✗
P4	Bounce-height decay	E		90.76 ✓	88.19 ✓	57.46 ✓	0.00 ✗	85.66 ✓	48.67 ✓	83.03 ✓	5.88 ✗
P5	Equal-mass collision	H		25.37 ✓	26.06 ✓	14.50 ✓	19.09 ✓	17.28 ✓	4.97 ✗	11.25 ✗	7.50 ✗
2. Rolling, Friction, and Rigid-Body Statics											
P6	Solid-sphere rolling	M		75.56 ✓	45.82 ✓	28.79 ✓	31.53 ✓	35.39 ✓	18.77 ✓	37.11 ✓	14.78 ✗
P7	Solid sphere vs. hoop	M		32.01 ✓	40.83 ✓	30.36 ✓	41.63 ✓	17.28 ✓	22.08 ✓	39.36 ✓	3.75 ✗
P10	Rough-incline round trip	H		24.43 ✓	28.84 ✓	8.10 ✗	12.36 ✗	14.43 ✗	19.18 ✗	9.89 ✗	0.00 ✗
P42	Mass-independent sliding	E		66.00 ✓	97.89 ✓	98.46 ✓	51.13 ✓	50.98 ✓	57.50 ✓	36.06 ✓	28.75 ✗
P43	Edge-pivot toppling	M		34.27 ✓	66.53 ✓	34.03 ✓	4.76 ✗	43.23 ✓	14.38 ✗	18.99 ✓	0.00 ✗
P44	Hanging-chain equilibrium	E		68.85 ✓	70.29 ✓	71.09 ✓	69.71 ✓	71.26 ✓	69.11 ✓	76.95 ✓	69.60 ✓
3. Pendulum Motion and Oscillations											
P8a	Small-angle isochronism	M		82.60 ✓	51.35 ✓	66.77 ✓	21.90 ✗	29.69 ✓	23.40 ✗	32.52 ✗	25.64 ✓
P8b	Pendulum period vs. mass	E		80.41 ✓	72.58 ✓	68.16 ✓	56.28 ✗	35.56 ✓	19.25 ✗	52.08 ✓	40.38 ✓
P8c	Large-angle pendulum	E		57.76 ✓	79.88 ✓	62.81 ✓	36.61 ✗	53.77 ✓	46.92 ✓	73.52 ✓	48.41 ✗
P9	Pendulum period vs. length	E		66.15 ✓	65.41 ✓	61.93 ✓	55.83 ✓	67.11 ✓	25.36 ✓	16.00 ✗	26.03 ✗
4. Optics and Projective Geometry											
P11	Light refraction	E		65.59 ✓	65.68 ✓	56.38 ✓	66.18 ✓	57.94 ✓	14.63 ✓	15.00 ✓	15.00 ✓
P12	Total-reflection critical angle	H		9.09 ✗	31.67 ✓	13.69 ✗	22.86 ✗	14.50 ✗	14.88 ✗	39.66 ✓	7.87 ✗
P13	Light reflection	M		96.15 ✓	49.82 ✗	0.00 ✗	95.75 ✓	0.00 ✗	3.38 ✗	24.73 ✗	0.00 ✗
P14	Projection concurrency	M		67.14 ✓	45.62 ✗	48.69 ✓	69.64 ✓	32.45 ✓	11.63 ✗	7.13 ✗	3.38 ✗
P16	Collinear-point cross-ratio	E		81.10 ✓	82.97 ✓	82.39 ✓	83.92 ✓	68.12 ✓	78.07 ✓	88.34 ✓	3.75 ✗
5. Hydrostatics and Buoyancy											
P18	Liquid-surface orientation	M		63.05 ✓	72.68 ✓	8.32 ✗	32.64 ✓	21.06 ✗	42.70 ✗	18.45 ✗	0.00 ✗
P19	Communicating vessels	E		97.01 ✓	98.16 ✓	97.70 ✓	99.01 ✓	93.09 ✓	97.40 ✓	95.44 ✓	94.05 ✓
P20	Floating-ice immersion	E		66.56 ✓	83.22 ✓	78.82 ✓	85.29 ✓	85.86 ✓	68.96 ✓	67.32 ✓	73.38 ✓
6. Phase Transitions and Melting											
P21	Ice melting: water level	H		95.61 ✓	0.00 ✗	15.00 ✓	0.00 ✗	0.00 ✗	3.75 ✗	0.00 ✗	0.00 ✗
P21b	Ice with a stone: melting	H		15.00 ✓	0.00 ✗	3.75 ✗	0.00 ✗	0.00 ✗	8.81 ✗	0.00 ✗	0.00 ✗
P21c	Freshwater ice in saltwater	H		46.87 ✓	0.00 ✗	57.50 ✓	0.00 ✗	0.00 ✗	0.00 ✗	0.00 ✗	14.38 ✗
P23	Freezing-induced expansion	M		94.30 ✓	96.21 ✓	66.69 ✗	3.38 ✗	0.00 ✗	86.43 ✓	3.75 ✗	0.00 ✗
P27	Crushed vs. intact ice	H		57.50 ✓	36.25 ✓	15.00 ✓	15.00 ✓	15.00 ✓	7.50 ✗	7.50 ✗	0.00 ✗
7. Electrostatics, Magnetism, and Electromagnetic Induction											
P28	Charged-sphere equilibrium	M		98.34 ✓	97.24 ✓	15.00 ✓	15.00 ✓	15.00 ✓	10.88 ✗	14.81 ✓	15.00 ✓
P34	Final compass orientations	M		36.43 ✓	15.05 ✓	15.00 ✓	18.01 ✓	36.32 ✓	16.41 ✓	15.00 ✓	15.95 ✓
P36	Eddy-current braking	E		36.25 ✓	89.38 ✓	78.75 ✓	50.00 ✗	75.00 ✓	57.50 ✓	53.75 ✓	14.38 ✗
P37	Closed vs. open jumping rings	M		23.00 ✓	20.84 ✓	15.00 ✓	15.00 ✓	36.25 ✓	57.50 ✓	15.00 ✓	15.00 ✓
P38	Solid vs. slotted plate damping	H		36.25 ✓	15.00 ✓	15.00 ✓	0.00 ✗	15.00 ✓	7.50 ✗	15.00 ✓	10.50 ✗
P39	Coil-induced light emission	E		57.50 ✓	57.50 ✓	68.12 ✓	57.50 ✓	57.50 ✓	68.13 ✓	32.50 ✗	3.75 ✗
8. Granular Media and Discharge Flow											
P40	Sandpile angle scaling	E		99.08 ✓	94.21 ✓	73.64 ✓	98.55 ✓	98.08 ✓	30.11 ✓	97.18 ✓	93.18 ✓
P41	Sand vs. water discharge	H		38.39 ✓	21.89 ✓	15.25 ✓	7.13 ✗	16.44 ✗	10.69 ✗	7.88 ✗	7.81 ✗
9. Surface Tension and Viscous Flow											
P45	Capillary rise vs. diameter	E		78.75 ✓	100.00 ✓	57.13 ✓	100.00 ✓	36.25 ✓	11.44 ✗	15.00 ✓	10.88 ✗
P47	Bubble-film curvature	M		15.00 ✓	55.08 ✓	25.30 ✓	28.79 ✓	42.26 ✓	32.41 ✓	42.62 ✓	15.00 ✓
P48	Droplet volume conservation	M		58.30 ✓	43.52 ✓	32.46 ✓	37.07 ✗	26.31 ✗	62.00 ✓	14.63 ✓	0.00 ✗
P49	Viscous settling speed	E		66.88 ✓	67.25 ✓	68.50 ✓	69.85 ✓	65.29 ✓	65.91 ✓	51.39 ✗	58.97 ✓
Average score				57.90	54.89	42.90	38.35	36.56	32.97	31.59	19.02

Table 2 Main results on PhysScope. Reviewed scores (0–100; higher is better) for 40 tasks. Bold marks row maxima, including ties at the displayed precision; backgrounds run from red (0) through yellow (50) to green (100). Within each *model-task pair*, ✓ means all available videos pass the consistency check; ✗ means at least one rejection. E/M/H denote Easy, Medium, and Hard, with 15, 15, and 10 tasks, respectively. Per-video scores use $S = 0.15C + 0.85P \mathbf{1}(F)$, where C is the observability score, P the physics score, and F the reviewed pass flag. Cells average available seeds; Average score is video-weighted.

Results by Difficulty Level. The empirical difficulty groups reveal differences in the physical constraints that current VDMs can reproduce reliably. Easy tasks include communicating-vessel equilibrium (P19), floating-ice immersion (P20), and hanging-chain equilibrium (P44). These tasks involve relationships that can be evaluated from relatively stable configurations, which may help explain their stronger performance. Medium tasks include free fall (P1) and pure rolling (P6), where reproducing recognizable motion is insufficient: the generated trajectories must also satisfy quantitative relationships, such as gravitational acceleration or the coupling between translation and rotation. For example, every model passes the consistency check on free fall, but the average score across models is only 26.61, indicating that evaluable videos can still depart substantially from the measured physical constraints. Hard tasks include equal-mass collisions (P5), rough-incline round trips (P10), and melting ice containing a stone (P21b). These scenarios require maintaining physical relationships through interactions or successive events, such as contact, motion reversal, and changes in object configuration. Their low scores are consistent with difficulties coordinating these processes over time. Limited evaluability also contributes: only 57.1% of Hard videos pass the consistency check, compared with 93.8% of Easy videos. Consequently, poor performance reflects difficulties both in producing sufficiently observable events and in satisfying the physical measurements. Model strengths also differ across these groups: MiniMax leads on Easy tasks, whereas Seedance achieves the highest Hard score of 37.76, substantially above MiniMax’s 19.87, despite remaining far from consistently accurate performance.

Analysis of Evaluator Groundedness. We assess evaluator groundedness using synthetic ground-truth (GT) videos with known physical relationships. These videos are generated as 2D animations using existing tools: object states are computed from prescribed physical relations and rendered frame by frame, while appearance variants preserve the underlying trajectories. We evaluate 480 correct GT videos across 40 tasks, with 12 videos per task. The VLM consistency gate is bypassed to assess the physical measurement module. As shown by the light-gray bars in Figure 2, GT videos obtain mean physical scores of 97.69, 96.90, and 98.25 on Easy, Medium, and Hard tasks, respectively, with an overall mean of 97.55 out of 100. These consistently high scores support the evaluator’s agreement with the known physical relationships encoded in the controlled scenes.

3.3 Agreement with Human Judgments

To assess whether automatic physical scores reflect human judgments, we compare our evaluator with a direct scoring baseline using Qwen3.6-27B (), referred to as Direct VLM. The baseline receives 24 sampled frames from each video, together with the original generation prompt and reference image, and directly predicts a physical correctness score in $[0, 1]$. Frames are resized to a maximum side length of 640 pixels, and inference uses a temperature of zero. Both methods are evaluated against the same human annotations on 320 videos across 40 tasks.

Score Correlation. Higher automatic physical scores should correspond to greater human-perceived physical correctness. To assess this alignment, we use individual-video ratings on a five-point scale, ranging from clear physical violation to no obvious violation. Videos with insufficient evidence for a physical judgment are marked separately. For each assessable video, we compute the mean of the available ratings included in the analysis:

$$H_i = \frac{1}{|\mathcal{R}_i|} \sum_{r \in \mathcal{R}_i} h_{i,r}, \quad (10)$$

where $h_{i,r} \in \{1, 2, 3, 4, 5\}$ and $\mathcal{R}_i$ contains the included annotators who provided a physical rating for video i . Since human and automatic scores use different scales, we quantify their alignment through within-task ranking agreement. Let $\mathcal{C}$ contain all unordered pairs of videos from the same task with both human and automatic scores available, including tied scores. We pool these pairs across tasks and compute:

$$A_{\text{rank}} = \frac{1}{|\mathcal{C}|} \sum_{(i,j) \in \mathcal{C}} \mathbf{1}[\text{sign}(P_i - P_j) = \text{sign}(H_i - H_j)], \quad (11)$$

where P_i denotes the automatic physical score and $\text{sign}(0) = 0$. Each pair is counted once, and agreement requires either the same strict ordering or a tie from both sources. Across all 1,120 within-task pairs, our evaluator achieves 52.23% agreement (585/1,120), compared with 46.07% (516/1,120) for Direct VLM, as shown in Fig. 3(a). Figure 3(b,c) additionally shows the individual-video score distributions. The reference $P = (H - 1)/4$ provides an illustrative linear mapping between the two scales and is used only for visualization.

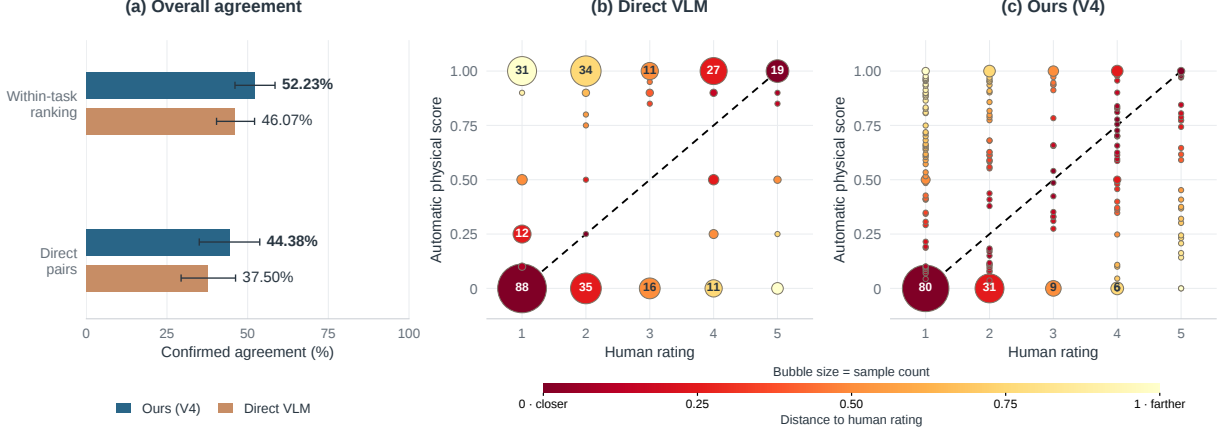


Figure 3 Agreement with human judgments. (a) Within-task ranking agreement and confirmed pairwise agreement, defined in Eqs. (11) and (12), with 95% task-bootstrap confidence intervals. (b,c) Automatic physical scores versus human ratings for Direct VLM (Qwen3.6-27B) and our evaluator, respectively. Bubble areas and labels indicate video counts. Black dashed lines show the illustrative reference $P = (H - 1)/4$; darker colors indicate smaller absolute deviations from this reference.

Pairwise Agreement. Direct comparisons provide a complementary assessment of whether the evaluator identifies the video that humans perceive as more physically correct, without deriving preferences from numerical ratings. Participants compare two videos from the same task and select either video, a tie, or insufficient evidence, focusing on physical correctness rather than visual quality. We aggregate assessable A/B/tie votes by strict majority, treating insufficient-evidence responses as abstentions. Let $\mathcal{Q}$ contain all predefined pairs with recorded automatic scores for both videos, and let $\mathcal{Q}_{\text{res}}$ contain those with a resolved majority judgment. For each resolved pair, $d_{a,b} \in \{-1, 0, +1\}$ represents a preference for video b , a tie, or a preference for video a , respectively. We compute confirmed pairwise agreement over the full set:

$$A_{\text{pair}}^{\text{conf}} = \frac{1}{|\mathcal{Q}|} \sum_{(a,b) \in \mathcal{Q}_{\text{res}}} \mathbf{1}[\text{sign}(P_a - P_b) = d_{a,b}], \quad (12)$$

where matching human and automatic ties count as agreement. Pairs without a resolved human judgment remain in the denominator but contribute no confirmed agreement; they are reported separately from disagreements. Among the 160 predefined pairs, our evaluator achieves 44.38% confirmed agreement (71/160), compared with 37.50% (60/160) for Direct VLM, as shown in Fig. 3(a). where matching human and automatic ties count as agreement. Pairs without a resolved human judgment remain in the denominator but contribute no confirmed agreement; they are reported separately from disagreements. Among the 160 predefined pairs, our evaluator achieves 44.38% confirmed agreement (71/160), compared with 37.50% (60/160) for Direct VLM, as shown in Fig. 3(a). Our evaluator achieves higher agreement on both measures, indicating that our metrics align more closely with human judgments of physical consistency.

3.4 Analysis of Prompt Refinement

4 Related Work

[Kaisen: Keep two Related Work paragraphs when restoring this section: Physics-Aware Video Generation and Video Generation Evaluation. Expand the first by approach: PhysGen uses rigid-body simulation, while PhysAnimator uses deformable-body simulation with sketch guidance (Liu et al., 2024; Xie et al., 2025). PhysCtrl guides video synthesis with generated 3D trajectories (Wang et al., 2025a); PhysDreamer transfers video dynamics priors to interactive 3D objects (Zhang et al., 2024). WISA encodes physical descriptions and properties, while VideoREPA aligns token relations with video foundation models (Wang et al., 2025b; Zhang et al., 2026). NewtonRewards constructs physics rewards from measurable proxies, and PhyWorld uses physics-preference optimization (Le et al., 2025; Zhao et al.,

2026). Close by explaining why these methods need task-specific tests of physical relationships. Retain the commented original while considering these additions.]

Video Generation Evaluation. Evaluating video generation requires assessing visual quality, temporal coherence, and semantic alignment. VBench (Huang et al., 2024) organizes these aspects into multiple dimensions, including subject consistency, motion smoothness, and spatial relationships, with evaluation protocols validated against human judgments. Physics-oriented benchmarks further examine whether generated events follow physical principles. VideoPhy (Bansal et al., 2025) evaluates physical commonsense in everyday activities, while PhyGenBench (Meng et al., 2024) introduces tasks spanning multiple physical domains and a hierarchical evaluation framework based on vision-language models. PhyGround (Lin et al., 2026) develops observable, law-specific criteria supported by human annotations and a specialized VLM evaluator. Complementary measurement-based approaches provide quantitative diagnostics: Principia (Thozhiyoor et al., 2026b) tests relational constraints between objects in Newtonian experiments, while GAUGE (Wang et al., 2026) uses measured trajectories, calibrated physical metadata, and task-specific observables to assess physical fidelity.

[**Kaisen:** Organize the evaluation paragraph by evidence: general video quality, VLM judgments, reference observations, and direct physical relationships. Add Physics-IQ Verified as the reference-based example (Rädsch et al., 2026). Keep GAUGE and Principia as distinct measurement approaches, then position PhysScope through multi-domain task-specific tests without implying that every earlier benchmark covers only mechanics.]

Building on these directions, our benchmark organizes controlled experiments across mechanics, optics, and other physical domains, grounding physical scores in explicit constraints on measurable quantities. This design supports interpretable, phenomenon-specific assessment without requiring a VLM to judge physical correctness.

5 Conclusion

We introduced PhysScope, a benchmark for evaluating physical consistency in video generation through 40 controlled tasks and explicit, task-specific measurements. Our evaluation framework separates temporal consistency and task observability from physical correctness, grounding scores in measurable relationships and retaining evidence for interpreting failures. Experiments across eight video generation models reveal persistent difficulties in reproducing quantitative physical relationships and completing the required physical events. Evaluation on synthetic reference videos supports the measurement module under controlled conditions, while human evaluation shows higher confirmed agreement than direct VLM scoring in both within-task rankings and pairwise comparisons. These findings highlight the value of explicit physical measurements for assessing generated videos. Current evaluation remains constrained by the visibility and reliable extraction of task-relevant evidence. Future work will extend task coverage and improve measurement robustness in more complex scenes. We hope PhysScope provides a useful foundation for diagnosing physical inconsistencies and developing video generation models that more reliably reproduce physical phenomena.

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A Detailed Task Catalog

PhysScope contains 40 tasks across nine physical families, with 65 evaluation metrics. Table 3 summarizes the task distribution. Each entry specifies the task definition, measured quantities, and score normalization.

Table 3 Overview of the nine physical families, their evaluation targets, and empirical difficulty distribution. E, M, and H denote Easy, Medium, and Hard.

Physical family	Task IDs	Physical targets	E	M	H
Translational motion and collisions	P1, P2, P3, P4, P5	Acceleration, projectile geometry, restitution, and momentum.	1	2	2
Rolling, friction, and rigid-body statics	P6, P7, P10, P42, P43, P44	Rolling, friction, supported rotation, and chain geometry.	2	3	1
Pendulum motion and oscillations	P8a, P8b, P8c, P9	Period relations for amplitude, mass, and length.	3	1	0
Optics and projective geometry	P11, P12, P13, P14, P16	Ray geometry, shadow concurrence, and projective invariance.	2	2	1
Hydrostatics and buoyancy	P18, P19, P20	Surface orientation, equilibrium levels, and submerged fraction.	2	1	0
Phase transitions and melting	P21, P21b, P21c, P23, P27	Level changes, freezing expansion, and comparative melting evidence.	0	1	4
Electricity, magnetism, and induction	P28, P34, P36, P37, P38, P39	Symmetry, needle directions, magnetic damping, and induction events.	2	3	1
Granular media and discharge flow	P40, P41	Repose-angle consistency and flow dependence on filling height.	1	0	1
Surface tension and viscous flow	P45, P47, P48, P49	Capillary ordering, film curvature, droplet volume, and settling speed.	2	2	0
Total	40 tasks	9 physical families	15	15	10

Notation and normalization. We denote metric residuals by e_j and normalized scores by $s_j \in [0, 1]$, with higher scores indicating better agreement. Unless stated otherwise, residuals are dimensionless. We use

$$q(e; a) = \frac{1}{1 + |e|/a}, \quad G(s_1, \dots, s_n) = \left(\prod_{j=1}^n s_j \right)^{1/n},$$

$$\text{CV}(x) = \frac{\text{std}(x)}{|\text{mean}(x)|}, \quad \text{clip}(x) = \min(1, \max(0, x)).$$

The scale a gives $q = 0.5$ at $|e| = a$; $\mathbb{I}[A]$ is one when A holds and zero otherwise. Image y increases downward. The physical score is the arithmetic mean of defined metric scores; tasks without M2 use M1 alone. Unless a task-specific rule applies, unresolved required measurements contribute zero while remaining unmeasured.

A.1 Translational Motion and Collisions

P1: Free fall (Medium).

Definition. A ball is released from rest in a fixed side view and falls under gravity before ground contact.

Metrics. M1 measures acceleration consistency, $e_1 = \text{CV}(\Delta v_y)$, using equal-interval velocity increments. M2 compares displacements d_1, d_2, d_3 in three equal intervals:

$$e_2 = \sqrt{\frac{(d_2/(3d_1) - 1)^2 + (d_3/(5d_1) - 1)^2}{2}}.$$

The first displacement must exceed localization noise; resolved constant-speed descent fails the acceleration test.

Normalization. $s_1 = q(e_1; 0.10)$ and $s_2 = q(e_2; 0.10)$.

P2: Oblique projectile motion (Hard).

Definition. A ball is launched at 45° , rises to its apex, and returns to its launch height in a fixed side view.

Metrics. M1 measures $e_1 = H/R - 1/4$, where H is apex height and R is range. M2 measures $\text{CV}(v_x)$, $\text{CV}(\Delta v_y)$, and parabola-fit error $u = \text{RMS}/R$. If return to launch height is absent but launch, apex, and descent are resolved, M1 uses $H/(2X_{\text{apex}}) - 1/4$ as a partial-arc estimate.

Normalization.

$$s_1 = q(e_1; 0.10), \quad s_2 = G(q(\text{CV}(v_x); 0.10), q(\text{CV}(\Delta v_y); 0.10), q(u; 0.05)).$$

P3: Complementary-angle projectiles (Medium).

Definition. Identical balls are launched simultaneously at 30° and 60° with equal initial speed, following separate lanes and landing at their launch heights.

Metrics. M1 measures range agreement, $e_1 = R_{30}/R_{60} - 1$. M2 combines the two parabola-fit RMS errors divided by ball diameter, u_{30}, u_{60} , and initial-speed mismatch $e_v = |v_{30}/v_{60} - 1|$.

Normalization.

$$s_1 = q(e_1; 0.15), \quad s_2 = G(q(u_{30}; 0.30), q(u_{60}; 0.30), q(e_v; 0.20)).$$

P4: Successive bounce heights (Easy).

Definition. A ball is dropped onto a horizontal hard surface and produces at least four visible rebounds with decreasing peak heights.

Metrics. For adjacent rebounds, define $r_h = h_{\text{next}}/h_{\text{prev}}$, $e_h^* = \sqrt{r_h}$, and $e_t^* = \Delta t_{\text{next}}/\Delta t_{\text{prev}}$. M1 compares height- and timing-based restitution and penalizes energy gain:

$$e_1 = \text{mean } |e_h^* - e_t^*|, \quad e_p = \text{mean } \max(0, e_h^* - 1).$$

M2 measures height increase $e_2 = \text{mean } \max(0, r_h - 1)$ and, when measurable, restitution variability $c = \text{CV}(e_{\text{restitution}})$.

Normalization. $s_1 = G(q(e_1; 0.18), q(e_p; 0.15))$. M2 is $s_2 = G(q(e_2; 0.15), q(c; 0.35))$ when c is measurable, and $s_2 = q(e_2; 0.15)$ otherwise. Trackable single motion without repeated rebounds receives $s_1 = s_2 = 0.1$.

P5: Equal-mass head-on collision (Hard).

Definition. A moving red ball strikes an initially stationary blue ball of equal mass on a level surface in a head-on elastic collision.

Metrics. M1 measures horizontal momentum agreement:

$$e_1 = \frac{v_{1,\text{after}} + v_{2,\text{after}}}{v_{1,\text{before}}} - 1.$$

M2 is not defined; elasticity is not separately scored.

Normalization. $s_1 = q(e_1; 0.10)$.

A.2 Rolling, Friction, and Rigid-Body Statics

P6: Pure rolling of a solid sphere (Medium).

Definition. A marked sphere is released without a push onto an incline and its horizontal extension, rolling while maintaining surface contact.

Metrics. M1 robustly fits center displacement against unwrapped rotation angle, yielding slope b and residual $e_1 = |b|/r - 1$. M2 measures

$$e_2 = \text{median} \frac{|v - \omega r|}{\max(v, \epsilon)}$$

over valid moving frames, where r is sphere radius.

Normalization. $s_1 = q(e_1; 0.25)$ and $s_2 = q(e_2; 0.35)$.

P7: Rolling solid sphere versus thin ring (Medium).

Definition. A solid sphere and a thin ring of equal outer radius roll without slipping from rest at the same height on one incline.

Metrics. M1 takes the median ring-to-sphere travel-time ratio r_t over common distances:

$$e_1 = \left| \frac{r_t}{\sqrt{10/7}} - 1 \right|.$$

M2 is not defined.

Normalization. $s_1 = q(e_1; 0.10)$.

P10: Sliding up and down a rough incline (Hard).

Definition. A block slides up an approximately 30° incline, stops, and slides down, with kinetic-friction coefficient 0.20.

Metrics. M1 compares ascent and descent acceleration magnitudes using the measured incline angle θ :

$$R_* = \frac{\sin \theta + 0.20 \cos \theta}{\sin \theta - 0.20 \cos \theta}, \quad e_1 = \left| \frac{a_{\text{up}}/a_{\text{down}}}{R_*} - 1 \right|.$$

M2 is not defined.

Normalization. $s_1 = q(e_1; 0.10)$. A confirmed ascent-only event receives $s_1 = 0.1$.

P42: Sliding onset independent of block mass (Easy).

Definition. A hinged board is gradually tilted until two blocks of different masses and matched contact conditions begin sliding.

Metrics. M1 compares onset frames, $e_1 = |f_l - f_r|/N_f$, where N_f is video length in frames. M2 compares onset inclinations, $e_2 = |\theta_{l,\text{onset}} - \theta_{r,\text{onset}}|$, in degrees.

Normalization. $s_1 = q(e_1; 1.0)$ and $s_2 = q(e_2; 90^\circ)$. Two measurable blocks that both remain stationary receive $s_1 = s_2 = 0.7$.

P43: Forced rotation about a support edge (Medium).

Definition. An actuator pushes a block's upper left face until it rotates about its lower right support edge and lies down without sliding away.

Metrics. M1 uses $E = \max(e_{\text{pivot}}, e_{\text{contact}}, e_{\text{shape}})$. Pivot drift and ground gap are their 95th-percentile values minus twice the pixel noise floor, clipped below at zero and divided by the initial block diagonal. Shape

error is noise-adjusted relative edge-length change. The observed rotation $\Delta\theta$ is the largest across continuous intervals of at least four frames and 0.3 s, without interpolating hidden poses. M2 is not defined.

Normalization. $s_1 = \mathbb{I}[\Delta\theta \geq 5^\circ]q(E; 0.02)$.

P44: Static shape of a hanging chain (Easy).

Definition. A chain with fixed equal-height endpoints is released from a nonequilibrium shape and settles under gravity.

Metrics. M1 measures catenary-fit error, $e_1 = \text{RMSE}_{\text{contour}} / \text{measured sag}$. M2 measures support-height difference divided by horizontal support span, $e_2 = \Delta h_{\text{support}} / L_{\text{span}}$.

Normalization. $s_1 = q(e_1; 0.10)$ and $s_2 = q(e_2; 0.02)$.

A.3 Pendulum Motion and Oscillations

P8a: Small-angle pendulum isochronism (Medium).

Definition. Two equal-length pendulums are released simultaneously from different small amplitudes and swing repeatedly.

Metrics. M1 compares periods, $e_1 = T_l / T_r - 1$. M2 fits a sinusoid to each angular trajectory and measures RMS residuals u_l, u_r in radians.

Normalization. $s_1 = q(e_1; 0.10)$ and $s_2 = G(q(u_l; 0.10 \text{ rad}), q(u_r; 0.10 \text{ rad}))$.

P8b: Pendulum period independent of mass (Easy).

Definition. Equal-length pendulums with different bob masses are released from the same angle at the same time.

Metrics. M1 compares periods, $e_1 = T_{\text{heavy}} / T_{\text{light}} - 1$. M2 compares first-frame lengths using pivots recovered from the swing trajectories:

$$e_2 = \frac{|L_l - L_r|}{\text{mean}(L_l, L_r)}.$$

Normalization. $s_1 = q(e_1; 0.10)$ and $s_2 = q(e_2; 0.10)$.

P8c: Finite-amplitude pendulum periods (Easy).

Definition. Equal-length pendulums are released from 15° and 30° ; the larger-amplitude pendulum should have a longer period.

Metrics. Both metrics use $r = T_{\text{large}} / T_{\text{small}}$. M1 measures $e_1 = |r - r_*|$; M2 measures the increase $r - 1$, where

$$r_* = \frac{K(\sin(30^\circ/2))}{K(\sin(15^\circ/2))} \simeq 1.0130520869$$

and K is the complete elliptic integral using the modulus convention.

Normalization. $s_1 = q(e_1; 0.05)$ and $s_2 = \text{clip}((r - 1) / (r_* - 1))$.

P9: Pendulum period versus length (Easy).

Definition. Identical pendulum bobs on strings with length ratio 1 : 2 are released from approximately equal small angles.

Metrics. M1 measures

$$e_1 = \left| \frac{(T_{\text{short}} / T_{\text{long}})^2}{L_{\text{short}} / L_{\text{long}}} - 1 \right|.$$

M2 measures cycle-to-cycle period variability, $c_s = \text{CV}(T_{\text{short}})$ and $c_l = \text{CV}(T_{\text{long}})$, over repeated oscillations.

Normalization. $s_1 = q(e_1; 0.20)$ and $s_2 = G(q(c_s; 0.20), q(c_l; 0.20))$.

A.4 Optics and Projective Geometry

P11: Refraction at an air–water interface (Easy).

Definition. A fixed laser illuminates a stationary air–water interface, with both incident and refracted rays visible.

Metrics. M1 measures Snell’s-law residual $e_1 = \sin \theta_i / \sin \theta_r - 1.333$, with angles relative to the normal. M2 compares ray–interface intersections p_i, p_r : $e_2 = \|p_i - p_r\| / W_c$, where W_c is calibrated container width.

Normalization. $s_1 = q(e_1; 0.12)$ and $s_2 = q(e_2; 0.025)$.

P12: Refraction and total internal reflection (Hard).

Definition. Four distinguishable beams change incidence directions at fixed points on a water interface, exhibiting refraction or total internal reflection.

Metrics. M1 measures the consistency of independently inferred refractive indices:

$$e_1 = \frac{\text{std}(n)}{\text{mean}(n)}.$$

Total internal reflection supplies a bound, not an exact critical-angle measurement. M2 is not defined.

Normalization. $s_1 = q(e_1; 0.05)$.

P13: Specular reflection (Medium).

Definition. A narrow laser beam strikes a plane mirror obliquely, with both rays and their contact point visible.

Metrics. M1 measures $e_1 = |\theta_i - \theta_r|$ in degrees. M2 measures $e_2 = \text{median}(\text{reflected-ray contact error}) / D_f$, where D_f is frame diagonal.

Normalization. $s_1 = q(e_1; 90^\circ)$ and $s_2 = q(e_2; 1.0)$.

P14: Concurrent shadows from a point source (Medium).

Definition. One stationary point light illuminates four vertical rods; backward extensions of their shadow axes should meet at a common point.

Metrics. M1 measures $e_1 = \text{RMS}_{\text{concurrency}} / D_f$, using shadow-axis line distances to their common intersection and frame diagonal D_f . M2 uses pairwise intersection estimates (x_k, y_k) :

$$e_2 = \frac{\text{mean}(\text{std}(x_k), \text{std}(y_k))}{D_f}.$$

Normalization. $s_1 = q(e_1; 0.05)$ and $s_2 = q(e_2; 0.05)$.

P16: Cross-ratio of four points on a rigid rod (Easy).

Definition. A rod with four fixed collinear markers slides with one endpoint on a wall and the other on the floor.

Metrics. For ordered projected marker positions $t_1, \dots, t_4$, M1 measures $e_1 = \text{CV}(\chi)$, where

$$\chi = \frac{(t_3 - t_1)(t_4 - t_2)}{(t_3 - t_2)(t_4 - t_1)}.$$

M2 measures $e_2 = \text{mean}(\text{RMS}_{\text{collinearity}} / \text{marker span})$ over frames.

Normalization. $s_1 = q(e_1; 0.10)$ and $s_2 = q(e_2; 0.05)$.

A.5 Hydrostatics and Buoyancy

P18: Liquid surface perpendicular to gravity (Medium).

Definition. A steel ball falls freely beside a vessel with a stationary liquid surface, providing a visible estimate of gravity direction.

Metrics. M1 measures $e_1 = \angle(\mathbf{a}_{\text{ball}}, \mathbf{t}_{\text{surface}}) - 90^\circ$. M2 measures surface line-fit RMS u_s and constant-acceleration trajectory-fit RMS u_b , both in pixels.

Normalization. $s_1 = q(e_1; 5^\circ)$ and $s_2 = G(q(u_s; 3 \text{ px}), q(u_b; 8 \text{ px}))$.

P19: Liquid levels in communicating vessels (Easy).

Definition. Liquid in a U-tube with unequal arm diameters settles after a disturbance, with no liquid added or removed.

Metrics. M1 measures final level difference, $e_1 = (\text{median}(y_l) - \text{median}(y_r))/H_{\text{ref}}$, where H_{ref} is the larger of frame height and the combined arm-region vertical span. M2 measures residual motion, $e_2 = \max(|b_l|, |b_r|)/H_{\text{frame}}$, using surface-position slopes against frame index. At least five overlapping terminal frames are required.

Normalization. $s_1 = q(e_1; 1.0)$ and $s_2 = q(e_2; 1 \text{ frame}^{-1})$.

P20: Submerged fraction of a floating ice column (Easy).

Definition. An intact vertical freshwater ice column reaches floating equilibrium without melting or touching its vessel.

Metrics. M1 measures the settled submerged-height ratio; M2 compares the left and right waterlines:

$$e_1 = \left| \text{median}\left(\frac{h_{\text{sub}}}{h_{\text{total}}}\right) - 0.917 \right|, \quad e_2 = \text{median} \frac{|y_l - y_r|}{W_{\text{vessel}}}.$$

Normalization. $s_1 = q(e_1; 0.10)$ and $s_2 = q(e_2; 0.05)$.

A.6 Phase Transitions and Melting

P21: Floating freshwater ice melting in freshwater (Hard).

Definition. Floating freshwater ice melts completely in a straight-sided freshwater vessel without material exchange; initial and final levels should agree.

Metrics. M1 measures level change; M2 measures vessel-width change:

$$e_1 = \frac{y_{\text{before}} - y_{\text{after}}}{H_{\text{vessel}}}, \quad e_2 = \frac{|W_{\text{after}} - W_{\text{before}}|}{W_{\text{before}}}.$$

Normalization. $s_1 = q(e_1; 0.05)$ and $s_2 = q(e_2; 0.05)$.

P21b: Melting floating ice containing a stone (Hard).

Definition. Floating ice melts and releases an embedded dense stone, which sinks; the final water level should decrease.

Metrics. M1 measures $e_1 = (y_{\text{before}} - y_{\text{after}})/H_{\text{initial}}$ after melting and a level change exceeding measurement uncertainty are observed. M2 is not defined.

Normalization. $s_1 = \mathbb{I}[e_1 < 0]$. Below-resolution changes remain unmeasured.

P21c: Freshwater ice melting in saltwater (Hard).

Definition. Freshwater ice melts and mixes with saltwater without material exchange; the final level should rise and no ice should remain.

Metrics. M1 records $e_1 = \Delta y / H_{\text{initial}}$, where $\Delta y = y_{\text{before}} - y_{\text{after}}$. M2 measures $r_s = E_{\text{solid,terminal}} / E_{\text{solid,initial}}$, an image-based solid-evidence ratio.

Normalization. $s_1 = \mathbb{I}[\Delta y > \delta_y]$, where δ_y is level-change uncertainty in pixels. M2 is $s_2 = \text{clip}(1 - r_s)$, overridden by $s_2 = 0$ if residual ice is confirmed in the final frame.

P23: Expansion when water freezes (Medium).

Definition. Water freezes completely in a straight-sided vessel without material transfer or a change in cross-section.

Metrics. M1 measures height expansion; M2 measures width stability:

$$e_1 = \frac{H_{\text{ice}}}{H_{\text{water}}} - \frac{1000}{917}, \quad e_2 = \frac{|W_{\text{final}} - W_{\text{initial}}|}{\max(W_{\text{initial}}, 1 \text{ px})}.$$

Normalization. $s_1 = q(e_1; 1.0)$ and $s_2 = q(e_2; 1.0)$.

P27: Crushed ice versus one ice block (Hard).

Definition. Equal masses of crushed ice and one intact ice block melt simultaneously in identical vessels.

Metrics. M1 compares liquid heights measured from each vessel's bottom and divided by its height. A qualifying lead occurs when the crushed-ice side exceeds the intact-ice side by more than their summed localization uncertainties. M2 is not defined.

Normalization. $s_1 = 1$ for a qualifying lead lasting at least 0.25 s, and $s_1 = 0$ for at least 0.50 s of continuous jointly readable evidence without such a lead. Otherwise the metric remains unresolved; gaps are not interpolated.

A.7 Electrostatics, Magnetism, and Electromagnetic Induction**P28: Symmetric equilibrium of charged balls (Medium).**

Definition. Two identical charged balls on equal insulating strings repel and settle into a symmetric outward-separated configuration.

Metrics. M1 measures string-angle asymmetry, $e_1 = |\theta_l - \theta_r|$, in degrees. M2 measures displacement asymmetry, $e_2 = |\ln(d_l/d_r)|$, using horizontal offsets from the apparatus centerline.

Normalization. $s_1 = q(e_1; 90^\circ)$ and $s_2 = q(e_2; 1.0)$. An observed failure of outward separation, settling, or fixed equal-length strings sets both scores to zero.

P34: Final directions of two compass needles (Medium).

Definition. A wire between two compasses is energized; their needles rotate and settle with opposing polar directions.

Metrics. M1 measures median smallest angular separation d in degrees over jointly readable observations among the final five frames, requiring at least two valid frames. M2 is not defined.

Normalization. $s_1 = (1 - \cos(\pi d/180))/2$; opposite directions score one and aligned directions score zero.

P36: Eddy-current braking on an incline (Easy).

Definition. Matched magnetic and nonmagnetic objects descend a copper incline; the magnetic object should arrive later and move more slowly.

Metrics. M1 measures $r_t = (f_m - f_0 + 1)/(f_c - f_0 + 1)$, where f_m, f_c are first-arrival frames at a common displacement and f_0 is the initialization frame. The common displacement is 80% of the smaller maximum displacement over jointly valid observations. M2 measures speed ratio $r_v = v_m/v_c$.

Normalization. $s_1 = \mathbb{I}[r_t > 1]$ and $s_2 = \mathbb{I}[r_v < 1]$.

P37: Closed versus split jumping rings (Medium).

Definition. Matched apparatuses simultaneously excite closed and split aluminum rings, each free to move along its fixed core.

Metrics. M1 measures $r_h = h_{\text{open}}/h_{\text{closed}}$, with each peak rise divided by that ring's initial outer diameter. A plateau or descent must establish each peak. M2 is not defined.

Normalization. $s_1 = \text{clip}((1 - r_h)/0.20)$. A reliably observed zero closed-ring rise scores zero, with the ratio undefined.

P38: Eddy-current damping of solid and slotted plates (Hard).

Definition. Solid and slotted conducting plates with matched mass and inertia swing through equivalent magnetic fields from the same release angle.

Metrics. M1 counts complete cycles $N_{\text{solid}}, N_{\text{slotted}}$ in a common window of at least 2 s, using cutoff $\max(1^\circ, 0.1 \text{ mean}(|\theta_{0,\text{solid}}|, |\theta_{0,\text{slotted}}|))$. Cycles are bounded by same-polarity events; truncated cycles are excluded. M2 is not defined.

Normalization. $s_1 = \mathbb{I}[N_{\text{solid}} < N_{\text{slotted}}]$. If $N_{\text{slotted}} = 0$, $s_1 = 0$ and the count ratio is undefined.

P39: Magnet entry and indicator illumination (Easy).

Definition. A magnet starts at rest, passes through a fixed coil connected to a bidirectional indicator, exits, and stops.

Metrics. M1 combines binary evidence of magnet entry E and confirmed indicator illumination during entry L , with a 0.5 s matching tolerance. An already-lit indicator qualifies if it remains lit during the matching interval. M2 is not defined.

Normalization. $s_1 = 0.5\mathbb{I}[E] + 0.5\mathbb{I}[L]$. Confirmed entry with an unreadable indicator receives the lower-bound score 0.5, with uncertainty range $[0.5, 1]$.

A.8 Granular Media and Discharge Flow**P40: Scale consistency of sand-pile repose angles (Easy).**

Definition. Two unequal-size piles of the same dry sand receive thin streams at their peaks and maintain comparable repose angles.

Metrics. M1 measures $e_1 = |\alpha_{\text{small}}/\alpha_{\text{large}} - 1|$. M2 measures left-right slope asymmetry in degrees:

$$e_2 = \max(|\alpha_{\text{small},l} - \alpha_{\text{small},r}|, |\alpha_{\text{large},l} - \alpha_{\text{large},r}|).$$

Normalization. $s_1 = q(e_1; 1.0)$ and $s_2 = q(e_2; 90^\circ)$.

P41: Discharge of water and sand from funnels (Hard).

Definition. Water and dry sand drain simultaneously from identical conical funnels with equal outlets and initial fill heights.

Metrics. M1 fits $Q \propto h^\beta$ for each material using cone geometry and an integrated volume/head fit:

$$e_1 = |\beta_{\text{water}} - 0.5| + |\beta_{\text{sand}}|.$$

M2 is not defined.

Normalization. $s_1 = q(e_1; 1.0)$.

A.9 Surface Tension and Viscous Flow**P45: Capillary rise in tubes of different radii (Easy).**

Definition. Two initially dry glass tubes of different inner radii are lowered into one liquid reservoir; liquid should rise higher in the narrower tube.

Metrics. M1 measures $d = \text{median}(h_{\text{narrow}} - h_{\text{wide}})$ over paired terminal observations, where $h = y_{\text{reservoir}} - y_{\text{meniscus}}$. M2 is not defined.

Normalization. $s_1 = \mathbb{I}[d > \delta_{\text{resolution}}]$, with d and measurement resolution in pixels.

P47: Curvature of the partition between unequal soap bubbles (Medium).

Definition. Two unequal near-spherical soap bubbles join and retain a visible separating film without rupture or detachment.

Metrics. M1 independently fits the outer radii r_s, r_l and signed partition radius r_p :

$$E_t = \left| r_p \left(\frac{1}{r_s} - \frac{1}{r_l} \right) - 1 \right|.$$

The residual e_1 is the frame-duration-weighted median of E_t over uniquely identifiable partition intervals of at least three frames and 0.08 s. M2 is not defined.

Normalization. $s_1 = q(e_1; 1.0)$. A resolved nearly straight partition whose curvature uncertainty excludes the prediction is assigned the unbounded-error limit, $s_1 = 0$.

P48: Volume conservation in droplet coalescence (Medium).

Definition. Two unequal free water droplets approach, touch, and merge into one near-spherical droplet.

Metrics. M1 measures $e_1 = |r_f^3/(r_1^3 + r_2^3) - 1|$ from fitted radii at a common image scale. M2 measures $e_2 = |\mathcal{R}_f - (\mathcal{R}_1 + \mathcal{R}_2)/2|$, where roundness is $\mathcal{R} = 1 - \text{RMS}_{\text{radial}}/r_{\text{fitted}}$.

Normalization. $s_1 = q(e_1; 0.10)$ and $s_2 = q(e_2; 0.10)$.

P49: Terminal settling of unequal spheres (Easy).

Definition. Two spheres of the same material and different radii settle through glycerin, reaching terminal speed before bottom contact.

Metrics. M1 measures

$$e_1 = \left| \frac{v_{\text{large}}/v_{\text{small}}}{(r_{\text{large}}/r_{\text{small}})^2} - 1 \right|.$$

Each speed is measured over the longest qualified nonzero terminal-speed window, with earlier windows preferred in ties; the radius ratio must be at least 1.1. M2 is not defined.

Normalization. $s_1 = q(e_1; 1.0)$.